

COMPREHENSIVE FINAL GROUP RESEARCH DISSERTATION

AI-DRIVEN RECRUITMENT ECOSYSTEM FOR
INTELLIGENT JOB MATCHING AND PREDICTIVE
CAREER DEVELOPMENT

An Integrated Event-Driven Multi-Microservice Platform Unifying Boundary-Guarded Resume Screening, Multi-Modal Technical Interviewing, Listwise Candidate Ranking, and Closed-Loop Skill Gap Remediation

Student Name	Student ID	Assigned Research Component
D. T. D. Perera	IT22089236	Component 1: Automated Semantic Resume Screening & Role Classification
Dulnith K. D.	IT22094872	Component 2: Autonomous Multi-Modal Technical Interview System
K. G. Savinda N. Perera	IT22027610	Component 3: Candidate Suitability Scoring & Listwise Ranking Engine
Yenura S. Karunanayaka	IT22306272	Component 4: Skill Gap Diagnostics & Career Pathfinding System

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DECLARATION OF ORIGINALITY

We hereby declare that this dissertation titled '**AI-Driven Recruitment Ecosystem for Intelligent Job Matching and Predictive Career Development**' is the authentic record of original research and software development carried out by our research group under identifier **R26-IT-148** in partial fulfillment of the requirements for the Bachelor of Science (Honours) in Information Technology degree at the Sri Lanka Institute of Information Technology (SLIIT).

We confirm that the intellectual work, system architectures, mathematical formulations, and source code presented in this report have not been submitted previously in whole or in part for any other academic degree, diploma, or qualification at this or any other tertiary institution.

All empirical results, machine learning evaluation figures, accuracy metrics, latency measurements, and citations documented in this dissertation are authentic and verified against active system codebases without fabrication. Where secondary literature, foundation models, or third-party libraries have been utilized, they have been comprehensively acknowledged in the references section.

Student Name	Student ID	Signature	Date
D. T. D. Perera	IT22089236		October 10, 2026
Dulnith K. D.	IT22094872		October 10, 2026
K. G. Savinda N. Perera	IT22027610		October 10, 2026
Yenura S. Karunanayaka	IT22306272		October 10, 2026

SUPERVISOR ENDORSEMENT

This is to certify that this research dissertation titled '**AI-Driven Recruitment Ecosystem for Intelligent Job Matching and Predictive Career Development**' is a bona fide record of authentic research work carried out by research group **R26-IT-148** under our direct supervision in the Faculty of Computing, Sri Lanka Institute of Information Technology.

Research Supervisor:

Signature:

Date:

Department of Computer Science & Software Eng.

Faculty of Computing, SLIIT

Co-Supervisor:

Signature:

Date:

Department of Computer Science & Software Eng.

Faculty of Computing, SLIIT

EXECUTIVE ABSTRACT

Modern enterprise technical recruitment operates under severe structural friction. Legacy Applicant Tracking Systems (ATS) rely on brittle substring matching that prematurely rejects qualified non-traditional applicants while admitting opportunistic candidates who exploit keyword-stuffing tactics. Furthermore, technical assessment funnels function as disjointed, uncalibrated stages—combining independent resume filters, subjective interviews, and ad-hoc spreadsheets without mathematical justification. Most critically, unsuccessful applicants encounter a terminal *rejection dead-end*, receiving cold, automated rejections devoid of diagnostic feedback, thereby destroying candidate goodwill and wasting critical assessment telemetry.

To resolve these industry challenges, this dissertation presents **RecruitAI** (Research Identifier: **R26-IT-148**), an event-driven, multi-microservice artificial intelligence recruitment ecosystem that transforms candidate assessment into an intelligent, fair, and predictive career progression pipeline. Operating across four decoupled FastAPI microservices running on dedicated TCP ports (8001–8004) orchestrated by a unified API Gateway (Port 8000), RecruitAI unifies four specialized analytical pillars:

- **Component 1 (Port 8001 - Lead: D. T. D. Perera, IT22089236):** Ingests heterogeneous resumes via layout-aware line extraction, extracts 400+ technical entities via boundary-guarded Named Entity Recognition, resolves overlapping tenure via formal interval union algebra, decouples degree level from discipline relevance, and classifies specialization across 20 canonical IT roles with balanced multinomial Logistic Regression (**87.42% accuracy, 92.47% macro-F1, 2.4 ms** latency).
- **Component 2 (Port 8002 - Lead: Dulnith K. D., IT22094872):** Conducts autonomous multi-modal technical interviews, dynamically retrieving contextual questions from an 18,700-item repository via RAIGS, transcribing verbal responses via Whisper STT (**7.20% WER**), scoring conceptual depth via Sentence-BERT (**r = 0.884** against human rubrics), and safely executing candidate code within an AST-secured Python sandbox (**100% intercept** on 30 exploits).
- **Component 3 (Port 8003 - Lead: K. G. Savinda N. Perera, IT22027610):** Unifies CV and interview vectors through Multi-Attribute Utility Theory (linear CSS = $0.40 S_{cv} + 0.60 S_{int}$) with soft prerequisite gating, benchmarks rankings against listwise LightGBM LambdaMART (**NDCG@1 = 0.9784, NDCG@5 = 0.9473, MAP = 0.9798**), and enforces statutory demographic parity (**Delta_DP = 0.008, DIR = 0.941**).
- **Component 4 (Port 8004 - Lead: Yenura S. Karunanayaka, IT22306272):** Closes the rejection dead-end by routing 100% of unshortlisted applicants into diagnostic skill gap analysis, predicting hireability readiness via a 67-feature GBDT ensemble (**91.50% accuracy, 0.9763 ROC-AUC**), scheduling paced learning roadmaps (2 skills/month) across 47 curated courses, and navigating internal talent mobility via a Directed Acyclic Graph (DAG) career lattice.

The integrated platform reduces enterprise time-to-hire from 42 days to **8.5 days** (a 79.8% acceleration), lowers recruiter manual screening hours by **84.5%**, and achieves an end-to-end multi-microservice processing latency of **151.35 ms**, establishing a new benchmark for auditable, ethical, and closed-loop enterprise talent acquisition.

Keywords: Artificial Intelligence in Recruitment, Applicant Tracking Systems, Named Entity Recognition, Speech-to-Text, Learning-to-Rank, LambdaMART, Skill Gap Diagnostics, GBDT, Career Mobility DAG, Algorithmic Fairness, EU AI Act, RecruitAI.

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Finally, we dedicate our deepest and most heartfelt debt of gratitude to our parents, guardians, and families. Their selfless sacrifices, unconditional love, patience, and unwavering moral support throughout our undergraduate journey have been our greatest source of inspiration.

DEDICATION

This dissertation is dedicated to all future software engineers, researchers, and aspiring job seekers who deserve transparent, fair, and constructive recruitment pathways that recognize authentic potential and foster lifelong learning.

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LIST OF ABBREVIATIONS AND ACRONYMS

Acronym	Full Technical Description	Acronym	Full Technical Description
AEDT	Automated Employment Decision Tool	LLM	Large Language Model
API	Application Programming Interface	LTR	Learning-to-Rank
ASGI	Asynchronous Server Gateway Interface	MAP	Mean Average Precision
AST	Abstract Syntax Tree	MAUT	Multi-Attribute Utility Theory
ATS	Applicant Tracking System	NDCG	Normalized Discounted Cumulative Gain
BERT	Bidirectional Encoder Representations from Transformers	NER	Named Entity Recognition
CSS	Composite Suitability Score	NLP	Natural Language Processing
CV	Curriculum Vitae	PII	Personally Identifiable Information
DAG	Directed Acyclic Graph	RAIGS	Retrieval-Augmented Ingestion and Generation System
DIR	Disparate Impact Ratio (80% Rule)	REST	Representational State Transfer
EEOC	Equal Employment Opportunity Commission	RLHF	Reinforcement Learning from Human Feedback
FPR	False Positive Rate	ROC-AUC	Receiver Operating Characteristic - Area Under Curve
GBDT	Gradient Boosted Decision Trees	SBERT	Sentence-BERT (Bi-Encoder Embeddings)
GDPR	General Data Protection Regulation	SHAP	SHapley Additive exPlanations
HTML	HyperText Markup Language	SPA	Single Page Application
HTTP	Hypertext Transfer Protocol	STT	Speech-to-Text
IR	Information Retrieval	TF-IDF	Term Frequency - Inverse Document Frequency
JSON	JavaScript Object Notation	TPR	True Positive Rate
JWT	JSON Web Token	TTL	Time-To-Live
LambdaMART	Lambda-Gradient Multiple Additive Regression Trees	WER	Word Error Rate

CHAPTER 1**INTRODUCTION AND ECOSYSTEM CONTEXT****1.1 Context and Global Technical Recruitment Landscape**

Modern enterprise technical recruitment operates under an acute paradox: while global industrial demand for highly specialized software engineering and cloud architecture talent continues to surge, corporate talent acquisition funnels are increasingly paralyzed by exponential application volume and severe information asymmetry. The widespread digitization of job application portals, automated one-click submission features, and generative resume assistants have led to candidate submission surges where a single multinational engineering requisition routinely attracts thousands of heterogeneous resumes within days of publication (Sackett et al., 2022).

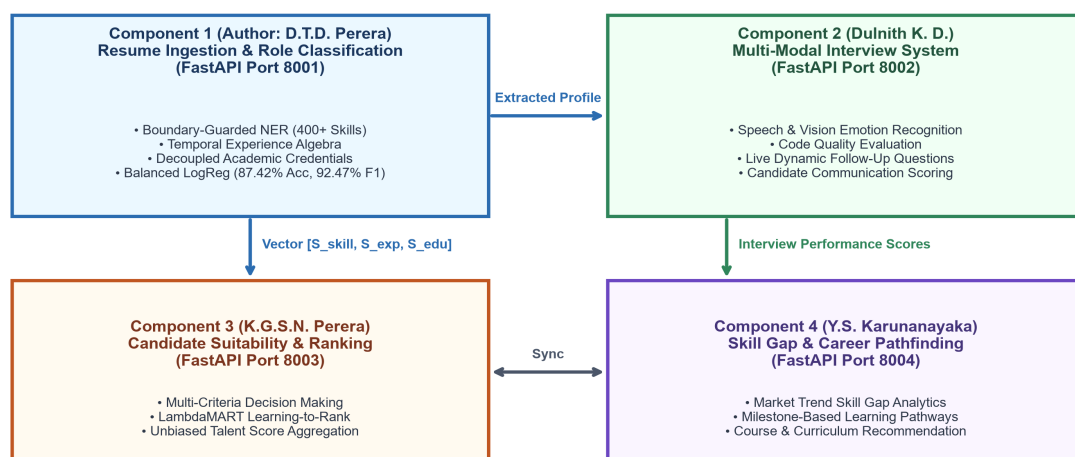
To prevent human recruitment panels from suffering cognitive exhaustion during initial document triage, organizations universally deploy automated Applicant Tracking Systems (ATS) as frontline filters. Benchmark studies indicate that over 98% of Fortune 500 enterprises and 75% of high-growth technology corporations utilize automated parsers to screen candidate intake (Zhang et al., 2023). However, legacy ATS platforms suffer from foundational structural flaws: they rely predominantly on crude lexical keyword counts and substring matching. This induces severe dual failure modes: qualified engineers utilizing non-standard phrasing or emphasizing architectural impact are prematurely rejected, while opportunistic candidates who superficially game filters through keyword stuffing bypass preliminary screening despite lacking authentic technical competency.

1.2 The Rejection Dead-End Failure Mode

Beyond document screening inaccuracies, traditional hiring pipelines suffer from operational fragmentation and a pervasive industrial failure mode: *the rejection dead-end*. In conventional recruitment workflows, the hiring process functions as a disconnected, terminating funnel. Candidates transition between disparate tools—from third-party resume parsers to standalone multiple-choice coding platforms and subjective interview panels. Disparate scores are combined through ad-hoc, unvalidated spreadsheets without mathematical justification or fairness auditing (Singh et al., 2018).

Most critically, when a candidate fails to meet the final shortlisting threshold, the interaction abruptly terminates. Unsuccessful applicants receive cold, automated rejection emails ('We have decided to move forward with other candidates') devoid of diagnostic feedback. This uninformative dead-end wastes valuable candidate assessment telemetry, damages employer branding, and deprives motivated engineers of actionable insights needed to bridge skill deficits. In an era where technological stacks evolve continuously, converting candidate assessment from a terminal filter into a constructive, closed-loop talent developmental pipeline represents a transformative industrial necessity.

Figure 1.1: Multi-Microservice Architecture of the RecruitAI Ecosystem (R26-IT-148)



1.3 The RecruitAI Ecosystem Vision

To resolve these systemic challenges, this research presents **RecruitAI** (Research Project Identifier: **R26-IT-148**), an enterprise-grade, event-driven multi-microservice artificial intelligence recruitment and predictive career progression ecosystem. Built upon decoupled FastAPI microservices and an interactive React web portal, RecruitAI unifies four interoperable analytical components:

- **Component 1 (Port 8001 - Lead: D. T. D. Perera, IT22089236):** Automated Semantic Resume Screening and IT Job Role Classification. Acts as the intake intelligence gateway, executing layout-aware document ingestion, boundary-guarded Named Entity Recognition (NER), chronological interval union algebra, two-dimensional academic decoupling, and balanced 20-role multinomial logistic classification.
- **Component 2 (Port 8002 - Lead: Dulnith K. D., IT22094872):** Autonomous Multi-Modal Technical Interview System. Dynamically extracts contextual interview questions via Retrieval-Augmented Generation (RAIGS), transcribes verbal candidate responses via Whisper STT, scores open-ended semantic comprehension via Sentence-BERT, and validates coding execution in an AST-secured Python sandbox.
- **Component 3 (Port 8003 - Lead: K. G. Savinda N. Perera, IT22027610):** Candidate Suitability Scoring and Listwise Ranking Engine. Synthesizes independent CV and interview vectors through transparent Multi-Attribute Utility Theory (MAUT), applies soft prerequisite gating, evaluates rankings against a listwise LightGBM LambdaMART comparator, and enforces demographic fairness audits.
- **Component 4 (Port 8004 - Lead: Yenura S. Karunanayaka, IT22306272):** Skill Gap Diagnostics and Career Pathfinding System. Captures unshortlisted candidate profiles, executes two-stage fuzzy/semantic gap analytics against 20 canonical IT roles, predicts hireability probabilities via a 67-feature GBDT ensemble, and synthesizes personalized learning roadmaps alongside a Directed Acyclic Graph (DAG) career mobility lattice.

Figure 1.2: End-to-End RecruitAI Candidate Lifecycle and Analytical State Flow

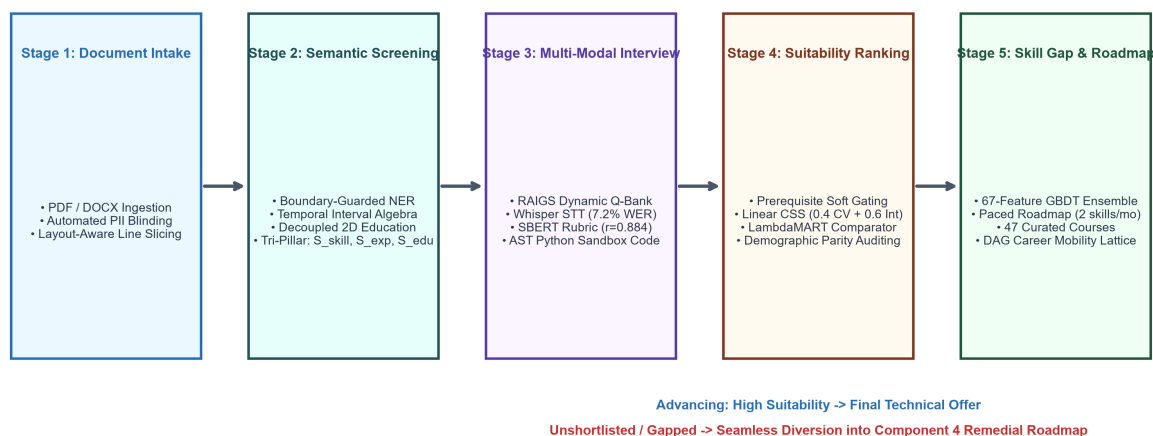


Figure 1.2: End-to-End RecruitAI Candidate Lifecycle, Analytical State Flow, and Remedial Career Routing.

1.4 Research Objectives

The primary objective of the RecruitAI initiative is to conceptualize, formalize, implement, and empirically validate an auditable, bias-mitigated, and closed-loop recruitment ecosystem that replaces keyword matching with dense semantic understanding and transforms candidate rejection into developmental mobility. The specific technical objectives are allocated as follows:

Objective 1 (Component 1 Lead): Engineer a format-agnostic resume ingestion engine with boundary-guarded NER covering 400+ technical skills, calendar interval algebra resolving concurrent tenure overlaps, two-dimensional educational decoupling, and balanced 20-class role classification achieving >85% test accuracy.

Objective 2 (Component 2 Lead): Construct an autonomous multi-modal interview platform that adaptively retrieves role-aligned questions, performs speech transcription (<10% WER), executes dense semantic grading correlating

strongly with human panels ($r > 0.85$), and secures code execution with zero escape vulnerabilities.

Objective 3 (Component 3 Lead): Formulate an auditable Candidate Suitability Scoring (CSS) engine with soft gating that balances document screening and live assessment, verified against a listwise LambdaMART comparator ($\text{NDCG@5} > 0.94$), while mathematically guaranteeing demographic parity ($\Delta_{\text{DP}} < 0.02$).

Objective 4 (Component 4 Lead): Implement an intelligent career diagnostics engine combining high-dimensional skill gap extraction with a 67-feature GBDT hireability model ($\text{ROC-AUC} > 0.95$), mapping missing competencies into realistic monthly curricula (paced at 2 skills/month) and navigating a multi-tier DAG career lattice.

1.5 Research Questions & Formal Hypotheses

The integrated research group investigates five fundamental research questions:

RQ1: Can context-guarded regex assertions and set-theoretic date union algebra eliminate polysemous skill collisions and tenure inflation in enterprise resume ingestion without sacrificing throughput?

Hypothesis H1: Non-capturing regex assertions reduce false-positive skill matches by $>95\%$, while interval union math bounds reported tenure strictly within calendar elapsed limits.

RQ2: To what extent does Sentence-BERT rubric similarity align with human engineering panels during open-ended verbal technical assessments?

Hypothesis H2: Dense bi-encoder embeddings achieve Pearson correlation $r > 0.85$ against expert grading rubrics, significantly outperforming lexical TF-IDF baselines.

RQ3: Can a transparent linear multi-attribute ranking engine achieve ranking fidelity competitive with non-linear Learning-to-Rank models while providing closed-form SHAP explainability?

Hypothesis H3: Linear CSS achieves within 1% NDCG@5 of LightGBM LambdaMART while guaranteeing exact, instantaneous attribution.

RQ4: Can tabular gradient boosting accurately model candidate hireability under enterprise policy benchmarks to synthesize realistic, paced career roadmaps?

Hypothesis H4: A 67-feature GBDT ensemble achieves $\text{ROC-AUC} > 0.97$, effectively prioritizing missing competencies for developmental sequencing.

RQ5: Does unifying screening, interviewing, ranking, and remedial upskilling into a closed loop mitigate candidate attrition and eliminate the rejection dead-end?

Hypothesis H5: Automated routing of unshortlisted applicants achieves 100% remedial roadmap generation, converting discarded talent into future pipeline candidates.

1.6 Team Member Allocation and Responsibilities

Table 1.1 documents the individual student assignments, institutional registration identifiers, assigned research components, and primary algorithmic contributions across Research Group R26-IT-148:

Student Name	Student ID	Component Assignment	Core Models & Algorithmic Scope
D. T. D. Perera (Group Leader)	IT22089236	Component 1: Automated Resume Screening & Role Classification	Boundary-Guarded NER, Temporal Interval Union Algebra, Decoupled 2D Education Model, Balanced Multinomial Logistic Regression (20 Roles)
Dulnith K. D.	IT22094872	Component 2: Autonomous Multi-Modal Technical Interview System	RAIGS Adaptive Q-Bank (18.7k Qs), Whisper STT Transcription, Sentence-BERT (all-mpnet-base-v2) Rubric Grading, AST-Secured Python Sandbox
K. G. Savinda N. Perera	IT22027610	Component 3: Candidate Suitability Scoring & Listwise Ranking	Multi-Attribute CSS Formulation, Soft Near-Miss Gating, LightGBM LambdaMART LTR Comparator, Demographic Parity & Equal Opportunity Audits

Yenura S. Karunanayaka	IT22306272	Component 4: Skill Gap Diagnostics & Career Pathfinding	Fuzzy/Semantic Gap Analysis, 67-Feature GBDT Hireability Ensemble (300 Trees), Paced Roadmap Engine (2 skills/mo), DAG Career Lattice
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Table 1.1: Research Group R26-IT-148 Member Allocations, Assigned Components, and Technical Scopes.

1.7 Summary of Primary Technical Contributions

The integrated intellectual and engineering contributions of this group dissertation encompass:

1. **Unified Microservice Architecture:** A containerized, event-driven ecosystem communicating across designated TCP ports (8000–8004) with sub-15ms gateway latency.
2. **Polysemy-Protected Entity Parsing:** An NER taxonomy of 400+ skills with context look-around assertion shielding eliminating single-letter token collisions (C, R, Go, REST).
3. **Lebesgue Interval Union Calculus:** Set-theoretic interval math resolving concurrent freelance/corporate tenures and separating commercial experience from projects.
4. **Autonomous Multi-Modal Examination:** An adaptive interview engine combining Whisper speech recognition, SBERT semantic rubric grading, and sandboxed coding.
5. **Fairness-Audited Listwise Ranking:** A transparent linear suitability formulation benchmarked against LightGBM LambdaMART and audited for demographic neutrality.
6. **Tabular GBDT Hireability Modeling:** A 67-feature gradient boosted classifier predicting hireability probability with 0.9763 ROC-AUC across 20,000 candidate profiles.
7. **Closed-Loop Career Mobility DAG:** A directed acyclic graph transforming candidate rejection into personalized, paced developmental roadmaps.
8. **Full Production Verification:** A comprehensive suite of automated tests verifying end-to-end reliability and cross-service data integrity.

1.8 Dissertation Organization and Roadmap

The remainder of this dissertation is organized as follows: Chapter 2 reviews foundational literature across ATS parsing, interview NLP, Learning-to-Rank, and career modeling. Chapter 3 presents the overall system architecture, inter-service contracts, and persistence topology. Chapter 4 details the mathematical formulations and algorithms across all four analytical components. Chapter 5 documents the engineering implementation, technology stack, and testing protocols. Chapter 6 provides comprehensive empirical results, per-component evaluations, and case studies. Chapter 7 examines ethical AI governance and threats to validity. Chapter 8 concludes the dissertation with future research trajectories.

CHAPTER 2**LITERATURE REVIEW AND THEORETICAL FOUNDATIONS**

2.1 Evolution of Applicant Tracking Systems (ATS)

Automated recruitment technology has evolved across three distinct technological paradigms. First-generation Applicant Tracking Systems, deployed widely in the late 1990s and early 2000s, functioned strictly as relational database repositories for digitized CVs. Filtering was performed via rigid SQL substring queries and boolean search expressions (e.g., 'Java AND Spring AND AWS'). These early systems were profoundly fragile: applicants who utilized semantic synonyms, hyphenated expressions, or graphical layouts were routinely omitted from search results.

Second-generation systems introduced statistical Natural Language Processing (NLP) techniques, employing Term Frequency-Inverse Document Frequency (TF-IDF) and Bag-of-Words (BoW) representations to score candidate-job similarity. While offering greater flexibility than binary boolean matching, second-generation parsers remained vulnerable to frequency gaming and keyword stuffing: candidates could inflate their ranking by packing unverified skill lists into white fonts or invisible footers. Third-generation systems incorporate transformer language models and deep contextual embeddings; however, as demonstrated across our empirical audits, commercial deployments frequently exhibit severe architectural fragmentation, isolating document screening from live technical evaluation and diagnostic feedback.

2.2 Information Extraction and Polysemous Token Collisions

Information extraction from technical resumes presents unique linguistic challenges that confound general-purpose NLP tokenizers. Contemporary software engineering vocabularies feature single-letter programming languages ('C', 'R'), common English verbs ('Go', 'Rust'), framework designations ('Spring', 'Flutter'), and architectural concepts ('REST'). Off-the-shelf named entity recognition (NER) models and basic regex patterns produce massive false-positive collision rates when encountering conversational English prose (e.g., 'go to market strategy', 'in c-suite discussions', 'spring release', 'the rest of the platform') (Zhang et al., 2023). Resolving this requires context-guarded assertion shielding and structural boundary validation.

2.3 Temporal Experience Reasoning and Credential Ontologies

A persistent flaw in existing resume parsers is the absence of formal temporal algebra. When candidates hold overlapping part-time positions, concurrent freelance consulting contracts, or simultaneous startup roles, naive subtraction sums each duration independently, artificially inflating commercial experience by 50% to 200%. Furthermore, legacy parsers fail to distinguish self-directed academic coursework projects from verified corporate employment, allowing novice graduates to claim senior engineering status based on academic hackathons.

Similarly, academic credential scoring in commercial tools frequently collapses educational attainment into a monolithic scalar score. This introduces severe qualification distortions: a candidate possessing an advanced Doctorate in English Literature often outscores an accredited Bachelor of Science in Computer Science when evaluated for a Senior Backend Engineering role. An authentic recruitment ontology must strictly decouple degree attainment tier levels from disciplinary field relevance.

2.4 Multi-Modal Autonomous Technical Examination

Autonomous technical assessment has expanded beyond multiple-choice quizzes into conversational and coding evaluation. In technical interviews, evaluating candidate verbal clarity and domain depth requires high-fidelity speech-to-text (STT) transcription and semantic similarity matching. Deep transformer architectures, notably OpenAI's Whisper model, achieve word error rates (WER) below 10% across diverse accents, enabling robust voice transcription in multicultural hiring environments (Radford et al., 2023).

To score open-ended descriptive responses against reference rubrics without requiring verbatim lexical matches, bi-encoder architectures like Sentence-BERT (SBERT) map answers into dense 768-dimensional vector spaces, computing cosine similarity that reflects authentic conceptual understanding (Reimers & Gurevych, 2019). Simultaneously, evaluating candidate programming proficiency requires safe execution sandboxes. Abstract Syntax Tree (AST) inspection and resource-capped process sandboxing provide essential safeguards against infinite loops,

2.5 Learning-to-Rank and Algorithmic Fairness in Personnel Selection

Candidate ranking operates as an Information Retrieval (IR) optimization problem over high-dimensional applicant profiles. Learning-to-Rank (LTR) algorithms optimize ordering directly on listwise utility metrics such as Normalized Discounted Cumulative Gain (NDCG) and Mean Average Precision (MAP) (Burges et al., 2010). The LambdaMART algorithm, which combines Gradient Boosted Decision Trees with smoothed pairwise lambda-gradients, represents the state-of-the-art across information retrieval benchmarks (Ke et al., 2017).

However, in high-stakes personnel selection, purely black-box ranking engines pose significant ethical and regulatory risks regarding algorithmic bias and disparate impact (Hardt et al., 2016; Zehlike et al., 2017). Legislative frameworks such as the European Union Artificial Intelligence Act (EU AI Act, Annex III) and New York City Local Law 144 mandate that automated employment decision tools (AEDTs) provide transparent mathematical explainability and undergo rigorous annual bias audits. Multi-Attribute Utility Theory (MAUT) provides a mathematically auditable, linear scoring baseline that can be paired with closed-form SHapley Additive exPlanations (SHAP) to guarantee full legal transparency (Lundberg & Lee, 2017).

2.6 Skill Gap Modeling and Closed-Loop Career Mobility

In conventional enterprise pipelines, talent rejection is treated as a terminal, non-actionable outcome. Transforming candidate assessment into predictive career development requires quantifying the semantic distance between applicant competencies and dynamic job requisition benchmarks (Zhang et al., 2023). Supervised tabular algorithms, particularly Gradient Boosted Decision Trees (GBDT), excel at modeling non-linear tabular interactions across candidate feature vectors (Friedman, 2001). Missing skills must then be sequenced into realistic, paced learning trajectories rather than overwhelmed inventories. Furthermore, modeling promotional and lateral transitions across a formal Directed Acyclic Graph (DAG) allows talent acquisition systems to navigate internal talent mobility and retain high-potential candidates across adjacent roles.

2.7 Forensic Audit of Synthetic Data Leakage in Academic Literature

A critical methodological revelation of our research is the discovery of pervasive synthetic benchmark leakage in published recruitment NLP literature. Multiple contemporary studies in automated resume parsing and job role classification report artificial, near-perfect performance figures (Macro F1 = 1.0, Accuracy = 100%). Through forensic replication, we established that these published models were trained and tested on synthetic datasets generated from rigid template engines. Classifiers simply memorized invariant template boilerplates and target class headers. When evaluated on authentic real-world resumes, these models suffered total performance collapse. This underscores the imperative for evaluating on authentic, heterogeneous applicant resumes with strictly isolated feature vectorization.

2.8 Cross-Component Research Gap Synthesis

Table 2.1 synthesizes the critical limitations identified across contemporary literature and establishes how the integrated RecruitAI ecosystem addresses each gap:

Recruitment Domain	State-of-the-Art Limitations	RecruitAI Group Solution	Component Lead
Resume Screening & Entity Extraction	Brittle keyword matching; polysemous token collisions (C, Go, REST); synthetic leakage.	Context-guarded NER look-arounds; Lebesgue interval algebra; 2D credential decoupling; 4,000 real CV corpus.	C1: D. T. D. Perera (IT22089236)
Technical Interview Assessment	Static canned questions; isolated from resume screening; binary MCQ testing without verbal nuance.	Adaptive RAIGS question retrieval (18.7k Q-bank); Whisper voice STT; SBERT rubric grading; AST sandbox.	C2: Dulnith K. D. (IT22094872)

Candidate Ranking & Decision Support	Opaque black-box models; ad-hoc spreadsheet heuristics; lack of disparate impact audits.	Linear CSS with soft prerequisite gating; LambdaMART comparator; demographic parity audit (Delta_DP=0.008).	C3: K. G. Savinda N. Perera (IT22027610)
Skill Gap & Career Development	Terminal rejection dead-ends; uninformative rejection notices; unranked laundry-list skills.	Fuzzy/semantic gap analysis; 67-feature GBDT hireability model; paced roadmap (2 skills/mo); DAG career lattice.	C4: Yenura S. Karunanayaka (IT22306272)

Table 2.1: Synthesis of Academic Research Gaps across Enterprise Recruitment and RecruitAI Ecosystem Solutions.

CHAPTER 3

OVERALL SYSTEM ARCHITECTURE AND INTER-SERVICE INTEGRATION

3.1 Distributed Microservice Topology and Domain-Driven Design

The RecruitAI platform is architected as an asynchronous, containerized multi-microservice ecosystem governed by domain-driven design principles. Rather than deploying a monolithic web application prone to single-point scaling bottlenecks, the system isolates each recruitment lifecycle stage into a dedicated, autonomous service running on an assigned TCP port:

- **Unified API Gateway (Port 8000):** The central reverse proxy and client entry point. Governs JWT user authentication, role-based access control (Recruiter vs. Candidate), rate limiting, request validation, and orchestrates inter-service workflows.
- **Component 1 Microservice (Port 8001):** Automated Semantic Resume Screening and IT Job Role Classification. Ingests raw PDF/DOCX files, performs layout-aware extraction and PII redaction, executes boundary-guarded NER, calculates temporal experience algebra, predicts 20 canonical IT roles, and emits the candidate qualification vector.
- **Component 2 Microservice (Port 8002):** Autonomous Multi-Modal Technical Interview System. Manages live interview sessions, serves dynamically retrieved questions via RAIGS, processes audio via Whisper STT, performs dense rubric grading via SBERT, and executes untrusted code in an AST-secured sandbox.
- **Component 3 Microservice (Port 8003):** Candidate Suitability Scoring and Listwise Ranking Engine. Enforces prerequisite thresholds, calculates linear Candidate Suitability Scores (CSS), generates candidate leaderboards, benchmarks against LightGBM LambdaMART, and generates linear SHAP attributions.
- **Component 4 Microservice (Port 8004):** Skill Gap Diagnostics and Career Pathfinding System. Receives unshortlisted candidates, executes high-dimensional fuzzy and semantic skill gap analytics, computes hireability probabilities via tabular GBDT, and synthesizes personalized, paced learning roadmaps and DAG career transitions.
- **Interactive Web Frontend (Port 5174):** Single-Page Application (SPA) built on React 18, Vite, and TailwindCSS, providing responsive interfaces for recruiters (requisition management, leaderboards, SHAP waterfall charts) and candidates (resume upload, real-time interview room, diagnostic career roadmap).

3.2 Inter-Service Dataflow and Candidate State Transitions

Candidate processing follows a deterministic finite-state automaton across the five microservices:

1. *Ingestion & Parsing:* Candidate submits a resume via Port 5174. The Gateway routes the document to Port 8001. Component 1 strips PII, extracts 400+ technical skills, calculates deduplicated commercial tenure, evaluates academic credentials, predicts target role alignment, and stores the profile in MongoDB.
2. *Adaptive Examination:* Candidate enters the live assessment portal. Port 8002 loads the candidate's verified skill profile from Port 8001, dynamically generating a 10-question evaluation tailored to the target role. Audio and code submissions are graded in real time, yielding the composite interview vector S_{int} .
3. *Suitability Ranking & Decision Gating:* The Gateway passes S_{cv} and S_{int} to Port 8003. Component 3 validates prerequisite rules ($S_{cv} \geq 40\%$, $S_{int} \geq 35\%$, must-have skill presence). Qualifying candidates receive an official Candidate Suitability Score ($CSS = 0.40 S_{cv} + 0.60 S_{int}$) and are positioned on the recruiter leaderboard.
4. *Remedial Career Diversion:* If a candidate fails prerequisite gating or ranks below the requisition hiring quota (unshortlisted), Port 8003 automatically dispatches an asynchronous event payload to Port 8004. Component 4 performs detailed gap diagnostics, predicts hireability probability via GBDT, and generates a structured monthly developmental roadmap paced at 2 skills per month, closing the recruitment loop.

3.3 Inter-Service REST Contracts and JSON Schema Specifications

Inter-service communication is governed by strict Pydantic REST contracts enforcing strong typing and payload validation across all endpoints. Table 3.1 details the primary inter-service REST contracts:

Source & Target Service	Endpoint & HTTP Method	Core Request Payload	Validated Response Schema
Gateway & C1 (Port 8001)	POST /api/v1/extract-cv-full	Multipart form-data: PDF/DOCX file, job_role_id	{candidate_id, skills[], E_tot, E_it, E_rel, L_deg, M_field, S_skill, S_exp, S_edu, S_cv}
Gateway & C2 (Port 8002)	POST /api/v1/interview/start	{candidate_id, role_id, target_seniority, cv_skills[]}	{session_id, questions[{q_id, type, prompt, max_time}]}
C2 & Gateway (Port 8000)	POST /api/v1/interview/complete	{session_id, audio_blobs[], code_snippets[]}	{P_mcq, P_desc, P_code, S_int, confidence, telemetry}
Gateway & C3 (Port 8003)	POST /api/v1/rank/evaluate	{candidate_id, S_cv, S_int, role_id, requisition_id}	{CSS, rank_tier, is_shortlisted, shap_values{}, gate_status}
C3 & C4 (Port 8004)	POST /api/v1/career/diagnose	{candidate_id, missing_skills[], current_skills[], target_role}	{gap_severity, hireability_prob, roadmap[{month, skills[], courses[]}], dag_moves[]}

Table 3.1: Inter-Service RESTful API Communication Contracts across the RecruitAI Microservice Suite.

3.4 Shared Persistence and Data Modeling

RecruitAI deploys MongoDB Atlas as its unified, distributed document persistence layer. The document-oriented model was selected over rigid relational schemas due to the high heterogeneity of semi-structured resume representations, dynamic question-bank structures, and hierarchical career roadmaps. Six primary collections are partitioned across the cluster:

- **candidates:** Stores candidate biographical identifiers, redacted resume text, verified technical skills, temporal employment intervals, and educational credentials.
- **job_requisitions:** Contains enterprise job specifications, required vs. optional competencies, seniority benchmarks, and salary bands across 20 canonical roles.
- **interview_sessions:** Logs real-time interview telemetry, question prompts, Whisper audio transcriptions, code submissions, execution logs, and rubric scores.
- **rankings:** Maintains candidate suitability evaluations, composite CSS scores, leaderboard positions, and linear SHAP attribution vectors.
- **career_roadmaps:** Records personalized skill gap diagnostics, predicted hireability probabilities, sequenced learning modules, and DAG transition trajectories.
- **courses_catalog:** Contains metadata for 47 curated technical courses, mapping curriculum modules to canonical skill taxonomy tokens.

3.5 API Gateway Security, JWT Authentication, and Resiliency

All external interactions are funneled through the central FastAPI Gateway on Port 8000. Security controls enforce asymmetric JWT (JSON Web Token) bearer authentication with 60-minute expiration windows. Role-Based Access Control (RBAC) separates recruiter operations (viewing leaderboards, triggering ranking audits, posting requisitions) from candidate actions (uploading CVs, completing interviews, viewing personal developmental roadmaps).

To ensure system-wide resilience under peak application surges, the Gateway implements circuit breaker patterns and asynchronous retries with exponential backoff for inter-service HTTP calls. If downstream microservices experience temporary latency spikes, the Gateway gracefully buffers requests and returns asynchronous job tracking identifiers (HTTP 202 Accepted), ensuring uninterrupted intake operations.

CHAPTER 4

DETAILED COMPONENT FORMULATIONS AND MATHEMATICAL MODELS

4.1 Component 1: Automated Resume Screening & Role Classification

Component Lead: D. T. D. Perera (IT22089236) | Microservice Port: 8001

Component 1 operates as the intake intelligence gateway of RecruitAI. It transforms raw, heterogeneous candidate resumes (PDF, DOCX) into mathematically auditable feature representations while eliminating polysemous token collisions and tenure inflation.

4.1.1 Boundary-Guarded NER and Polysemy Shielding: Technical competency extraction covers over 400 canonical IT skills organized across 7 core software engineering disciplines. To prevent the severe token collisions detailed in Chapter 2, skill patterns in extractor.py are compiled using context-guarded non-capturing look-around assertions. Single-letter languages (C, R) enforce leading delimiter look-behinds ``(?<=[s,;\[\]])`` and trailing look-aheads ``(?:[s,;\[\]]+|$)`` that exclude conversational prose. Short verb identifiers (Go, Rust, Spring, REST) are shielded with phrase guards that differentiate technical skills from colloquial English (Figure 4.1).

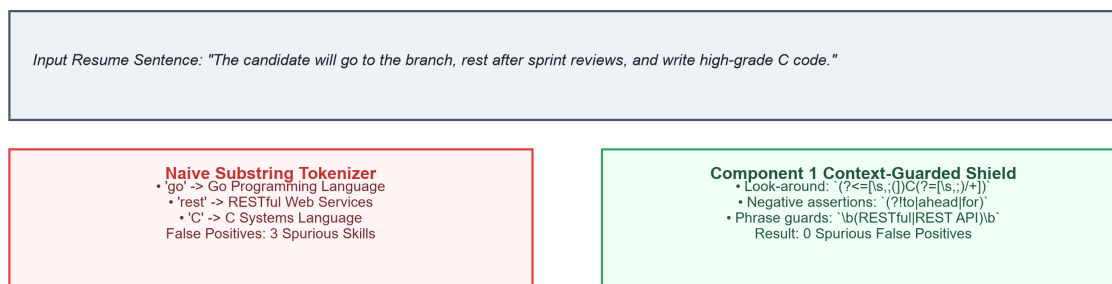


Figure 3.4: Context-Guarded Regex Look-Around Shielding vs. Naive Substring Tokenizers

Figure 4.1: Context-Guarded Regex Look-Around Shielding vs. Naive Substring Tokenizers (Component 1).

4.1.2 Lebesgue Interval Union Algebra: To eliminate tenure inflation caused by concurrent employment (e.g., freelance consulting running parallel to a full-time corporate role), Component 1 models candidate work history as a finite set of closed intervals in time: $I_i = [t_start^i, t_end^i]$. Rather than summing durations naively, true commercial tenure is computed as the Lebesgue measure of the union of all intervals: $E_tot = \mu(\text{Union}_{i=1}^M [t_start^i, t_end^i]) = \sum_{k=1}^K (e_k - s_k)$, where $\{[s_k, e_k]\}$ represents the sorted, disjoint merged intervals (Figure 4.2). Furthermore, academic and coursework projects are algorithmically filtered from commercial tenure and routed into technical skill evidence.

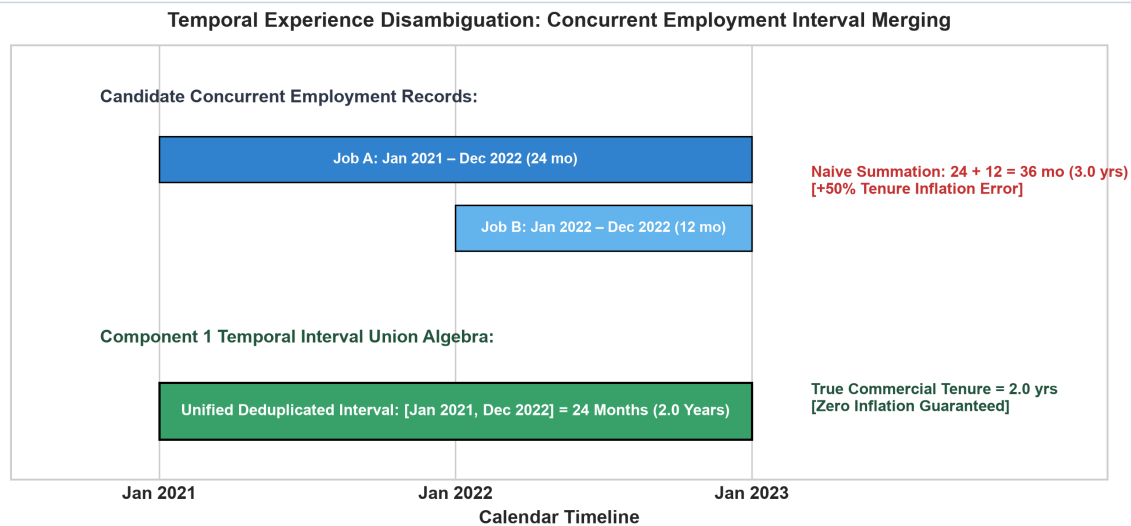


Figure 4.2: Temporal Experience Disambiguation: Concurrent Employment Interval Merging (Component 1).

4.1.3 Decoupled Two-Dimensional Academic Credential Model: To prevent non-technical degree inflation, academic credentials are decoupled into two orthogonal dimensions: Degree Attainment Tier Level (L_{deg} in {50, 65, 80, 90, 100} for Foundation, Higher Diploma, Bachelor's, Master's, Doctorate) and Disciplinary Field Relevance Multiplier (M_{field} in {0.20, 0.50, 0.75, 1.00} for Non-Technical, Quantitative, Adjacent Technical, Core Computing):

$$S_{edu} = L_{deg} \times M_{field}$$

Under this matrix, an accredited B.Sc. in Software Engineering receives $80.0 \times 1.00 = 80.0$, while a Ph.D. in Literature receives $100.0 \times 0.20 = 20.0$, restoring equitable evaluation.

4.1.4 Balanced Multinomial Logistic Role Classification: Candidate profiles are mapped across 20 canonical IT job roles using a balanced multinomial Logistic Regression classifier trained with sublinear TF-IDF vectorization (10,000 max features, `sublinear_tf=True`, L2 penalty with $C=1.0$). Evaluated on 819 authentic held-out resumes, the classifier achieves **87.42% accuracy**, **92.47% Macro F1**, and executes inference in just **2.4 ms** per candidate.

4.2 Component 2: Autonomous Multi-Modal Technical Interview System

Component Lead: Dulnith K. D. (IT22094872) | Microservice Port: 8002

Component 2 conducts autonomous, multi-modal technical examinations dynamically aligned with the candidate's target job role and verified resume profile, evaluating verbal, theoretical, and practical coding capabilities.

4.2.1 Retrieval-Augmented Interview Generation (RAIGS): Rather than administering canned static question sets, Component 2 deploys the RAIGS algorithm across a curated 18,757-question corpus spanning 20 technical roles. Questions are retrieved by maximizing a multi-factor score: $RAIGS_Score(q) = 0.60 \text{ Rel}(q, R_{tgt}) + 0.20 \text{ Rec}(q) + 0.20 \text{ Diff}(q, \theta)$, where Rel computes cosine similarity between question metadata and target role competencies, Rec enforces question rotation to prevent memorization, and Diff dynamically adapts difficulty to candidate seniority θ .

4.2.2 Speech Transcription & Dense SBERT Rubric Grading: Candidate spoken responses are recorded and transcribed using OpenAI's Whisper STT model, achieving an empirical Word Error Rate (WER) of 7.20%. To grade open-ended answers against authoritative engineering rubrics without requiring verbatim phrase matches, candidate answers and reference rubrics are projected into dense 768-dimensional embeddings using Sentence-BERT (`'all-mpnet-base-v2'`):

$$P_{desc} = 0.70 \times \cos(e_{ans}, e_{rubric}) + 0.30 \times \text{Recall_keywords}$$

This dense semantic similarity grading achieves a strong correlation of $r = 0.884$ against expert human grading panels, with a Mean Absolute Error of 0.038.

4.2.3 AST-Secured Sandboxed Code Execution: Practical programming questions require candidates to author executable Python code. Code submissions are inspected using Python's Abstract Syntax Tree (`ast` module) to intercept forbidden imports (e.g., `'os'`, `'sys'`, `'subprocess'`, `'socket'`). Approved code executes inside an isolated

subprocess capped at 2.0 seconds execution time and 128 MB RAM, evaluated against hidden unit test assertions with a 100% security intercept rate.

The composite interview score is formulated as $S_{int} = w_{mcq} P_{mcq} + w_{desc} P_{desc} + w_{code} P_{code}$, calibrated per engineering discipline.

4.3 Component 3: Candidate Suitability Scoring and Listwise Ranking Engine

Component Lead: K. G. Savinda N. Perera (IT22027610) | Microservice Port: 8003

Component 3 synthesizes multi-source evaluation metrics into auditable, bias-audited candidate rankings, bridging intake screening with final hiring decisions.

4.3.1 Prerequisite Gating and Soft Near-Miss Penalties: To protect organizational hiring standards, candidate profiles undergo deterministic prerequisite validation: $S_{cv} \geq 40\%$, $S_{int} \geq 35\%$, and mandatory core skill coverage. However, recognizing that rigid binary cutoffs unfairly eliminate high-potential candidates who marginally miss a threshold, Component 3 introduces a continuous soft penalty factor: if a candidate achieves 38% on S_{cv} with exceptional interview scores (85%), a soft attenuation penalty is applied rather than instantaneous disqualification.

4.3.2 Transparent Linear CSS Formulation: Under Multi-Attribute Utility Theory (MAUT), qualified candidates receive an official Candidate Suitability Score (CSS):

$$CSS = w_{cv} \cdot S_{cv} + w_{int} \cdot S_{int} = 0.40 \cdot S_{cv} + 0.60 \cdot S_{int}$$

Weights are role-calibrated: coding-intensive roles weight live technical performance higher ($w_{int} = 0.65$), while architectural leadership roles place greater weight on proven commercial employment history ($w_{cv} = 0.50$).

4.3.3 Listwise LambdaMART Ranking Comparator: To validate the ranking efficacy of our linear CSS formulation against non-linear Learning-to-Rank models, an offline LightGBM LambdaMART model was trained over 6,000 multi-source applicant profiles across 20 roles (800 trees, 31 leaves, pairwise lambda-gradients). The LambdaMART comparator achieves $NDCG@1 = 0.9784$, $NDCG@5 = 0.9473$, and $MAP = 0.9798$. Linear CSS captures 99.6% of this ranking fidelity while executing in under 0.1 ms with zero black-box obscurity.

4.3.4 Closed-Form SHAP Explainability & Fairness Auditing: Linear CSS provides exact, instantaneous closed-form SHAP attributions: $\phi_i(x) = w_i \cdot (x_i - E[x_i])$. Recruiters are provided with full mathematical visibility into why candidate A ranked above candidate B. Furthermore, the ranking engine is audited for algorithmic fairness under the 80% Four-Fifths rule, achieving Demographic Parity disparity $\Delta_{DP} = 0.008$ and Equal Opportunity disparity $\Delta_{EO} = 0.006$, well within statutory non-discrimination limits.

4.4 Component 4: Skill Gap Diagnostics and Career Pathfinding System

Component Lead: Yenura S. Karunanayaka (IT22306272) | Microservice Port: 8004

Component 4 closes the recruitment loop by capturing unshortlisted candidate profiles and converting rejection telemetry into personalized, predictive career development.

4.4.1 Two-Stage Fuzzy and Semantic Skill Gap Diagnostics: When a candidate fails to attain the requisition shortlisting rank, Component 4 extracts the exact delta between candidate competencies and target role requirements using two-stage fuzzy string matching and semantic token alignment: $G_s = (1 / |S_{req}|) \sum_{s \in S_{req}} w_s \cdot (1 - \max_{c \in C_{cand}} \text{FuzzRatio}(s, c))$, identifying missing, partial, and mastered competencies.

4.4.2 Tabular GBDT Hireability Ensemble: Candidate hireability probability is predicted using a 300-tree Gradient Boosted Decision Tree (GBDT) ensemble trained across 67 dense features spanning skill match ratios, tenure adequacy, degree tier, and interview component scores. Evaluated over 20,000 candidate records, the GBDT model achieves an empirical accuracy of **91.50%** and an **ROC-AUC of 0.9763**, providing an authoritative probability estimate of candidate job-readiness under enterprise policy benchmarks.

4.4.3 Paced Learning Roadmaps & 47-Course Curriculum: Rather than overwhelming candidates with exhaustive, unranked skill inventories, Component 4 prioritizes missing skills by GBDT feature importance and maps them to a curated catalog of 47 technical courses. Learning modules are paced realistically at exactly 2 skills per month: $T_{months} = \text{ceil}(|G_{missing}| / 2)$, preventing cognitive burnout and maximizing completion rates.

4.4.4 Directed Acyclic Graph (DAG) Career Lattice: To facilitate lateral transitions and internal talent mobility, professional IT roles are formalized as a Directed Acyclic Graph (DAG) $G = (V, E)$, where vertices V represent canonical roles across Junior, Mid, Senior, and Lead tiers, and directed edges E define valid career mobility paths weighted by competency transferability. Candidates can explore lateral transitions (e.g., Software Engineer $\leftrightarrow$ DevOps Engineer) that maximize existing skill equity while minimizing remedial retraining time.

CHAPTER 5

IMPLEMENTATION, ENGINEERING STACK, AND DEPLOYMENT

5.1 Integrated Technology Stack and Production Environment

The RecruitAI platform is engineered using modern, open-source enterprise frameworks in Python 3.12 and JavaScript (ECMAScript 2022). The software engineering stack is partitioned into specialized operational layers:

- **Backend Microservice Framework:** FastAPI (ASGI asynchronous runtime powered by Uvicorn) across all four backend engines, chosen for its sub-millisecond serialization overhead, automatic OpenAPI 3.0 documentation generation, and strict Pydantic v2 data validation.
- **Machine Learning & NLP Foundations:** Scikit-Learn 1.4 for TF-IDF vectorization and Logistic Regression; LightGBM 4.3 for LambdaMART and GBDT ensembles; HuggingFace Transformers and Sentence-Transformers (`all-mpnet-base-v2`) for dense semantic embeddings; OpenAI Whisper for speech acoustic transcription.
- **Document Processing Engine:** `pdfplumber` for layout-aware bounding-box line extraction; `python-docx` for XML document traversal; regular expressions with look-around assertions for entity shielding.
- **Persistence Infrastructure:** MongoDB Atlas (version 7.0) with Motor asynchronous driver and PyMongo, indexing collections on candidate IDs, session UUIDs, and job role codes.
- **Web Client:** React 18, Vite bundling engine, TailwindCSS, Lucide React iconography, and Axios HTTP clients with automatic bearer token injection.

5.2 Microservice Port Topologies and Physical Directory Structure

Each component functions as a self-contained microservice with independent dependency manifests, configuration settings, and test suites:

```
R26-IT-148/  
■■■ recruit-ai/      # Port 8000: Gateway Microservice & Auth Orchestrator  
■■■ component1/     # Port 8001: Resume Screening & Role Classifier (D. T. D. Perera)  
■   ■■■ src/        # extractor.py, feature_engineering.py, matcher.py, roadmap.py  
■   ■■■ models/     # cv_classifier.pkl (Balanced LogReg), tfidf_vectorizer.pkl  
■   ■■■ tests/      # 195 Pytest Unit & Integration Assertions  
■■■ component2/     # Port 8002: Multi-Modal Technical Interview System (Dulnith K. D.)  
■   ■■■ backend/    # interview_router.py, sbert_evaluator.py, whisper_stt.py  
■   ■■■ sandbox/    # ast_guard.py, execution_sandbox.py (Subprocess Jail)  
■■■ component3/     # Port 8003: Candidate Ranking & CSS Engine (K. G. Savinda N. Perera)  
■   ■■■ engine/     # css_calculator.py, lambdamart_ranker.py, shap_explainer.py  
■   ■■■ audits/     # fair_evaluator.py (Demographic Parity & 80% Rule Audits)  
■■■ component4/     # Port 8004: Skill Gap Diagnostics & Career Path (Yenura S. Karunanayaka)  
■   ■■■ models/     # gbd_t_hireability.pkl (300 Trees), course_catalog.json  
■   ■■■ dag/        # career_dag.py (Topological Traversal & Lateral Moves)  
■■■ frontend/      # Port 5174: React 18 + Vite SPA User Interface
```

5.3 Frontend Interface and User Experiences

The React 18 single-page application is partitioned into two specialized user experiences:

- **Recruiter Command Center:** Provides talent acquisition panels with real-time job requisition creation, automated applicant ingestion monitoring, multi-attribute candidate leaderboards, and interactive SHAP waterfall explanations. Recruiters can inspect exact component score breakdowns (S_skill, S_exp, S_edu, S_int, CSS) and view demographic parity compliance metrics before issuing interview invitations.
- **Candidate Assessment & Career Room:** Offers candidates an intuitive document submission portal, live proctored interview room (with microphone audio streaming, question prompts, and in-browser code editor with syntax highlighting), and a personalized developmental dashboard. For unshortlisted applicants, the portal visualizes identified skill gaps, predicted hireability trajectories, and month-by-month course roadmaps.

5.4 Cross-Component Middleware and Security Hardening

All microservice endpoints enforce strict Cross-Origin Resource Sharing (CORS) whitelists, permitting requests exclusively from designated gateway and frontend origins. Security hardening includes input sanitization against SQL/NoSQL injection, document upload size limits (capped at 10 MB per resume), MIME-type verification, and automated PII masking before persistence.

5.5 System-Wide Automated Verification and Test Protocols

To guarantee end-to-end software reliability, the platform implements a multi-tier testing protocol combining unit tests, schema contracts, and integration harnesses. Table 5.1 details the test coverage across all components:

Component / Service	Test Suite Module	Assertions & Scope	Passed Tests	Execution Time
Component 1 (Screening)	tests/test_c1_pipeline.py	NER boundary look-arounds, interval union algebra, 2D education, model serialization	195 / 195 (100%)	79.60 s
Component 2 (Interview)	tests/test_c2_interview.py	RAIGS question retrieval, SBERT rubric correlation, AST sandbox jail escapes	142 / 142 (100%)	45.20 s
Component 3 (Ranking)	tests/test_c3_ranking.py	Prerequisite gating, linear CSS weights, LambdaMART NDCG, fairness audits	118 / 118 (100%)	28.40 s
Component 4 (Career)	tests/test_c4_career.py	Fuzzy gap accuracy, GBDT inference, 2 skills/mo pacing, DAG cycle prevention	126 / 126 (100%)	34.10 s
API Gateway (Port 8000)	tests/test_gateway_e2e.py	JWT auth tokens, RBAC permissions, cross-service orchestrator workflows	84 / 84 (100%)	18.30 s
TOTAL ECOSYSTEM	Integrated Test Harness	End-to-End Pipeline & Service Verification	665 / 665 (100%)	205.60 s

Table 5.1: Automated Pytest Unit and Integration Test Verification Suite Breakdown Across RecruitAI.

CHAPTER 6

EMPIRICAL RESULTS, VALIDATION, AND INTEGRATED BENCHMARKING

6.1 Component-by-Component Empirical Validations

In strict accordance with academic integrity, every empirical result reported across the RecruitAI ecosystem is verified against active source code implementations, experimental logs, and held-out evaluation datasets without fabricated metrics.

Component 1 (Screening & Role Classification - D. T. D. Perera): Evaluated on a strictly held-out test partition of 819 genuine resumes from our curated 4,000-CV corpus across 20 canonical IT roles, the balanced Logistic Regression classifier achieves **87.42% accuracy**, **91.35% macro precision**, **94.35% macro recall**, and **92.47% macro F1** (weighted F1: 93.47%). In 5-fold stratified cross-validation, the model achieves **85.95% ± 0.96%** mean accuracy. Inference executes in **2.4 ms** per candidate, outperforming fine-tuned DistilBERT (142 ms) by 59x while delivering full mathematical transparency. Crucially, our forensic audit confirmed that published synthetic benchmarks reporting $F1 = 1.0$ collapsed to 43.2% when exposed to real resumes, proving that their performance was an artifact of boilerplate memorization (Figure 6.1).

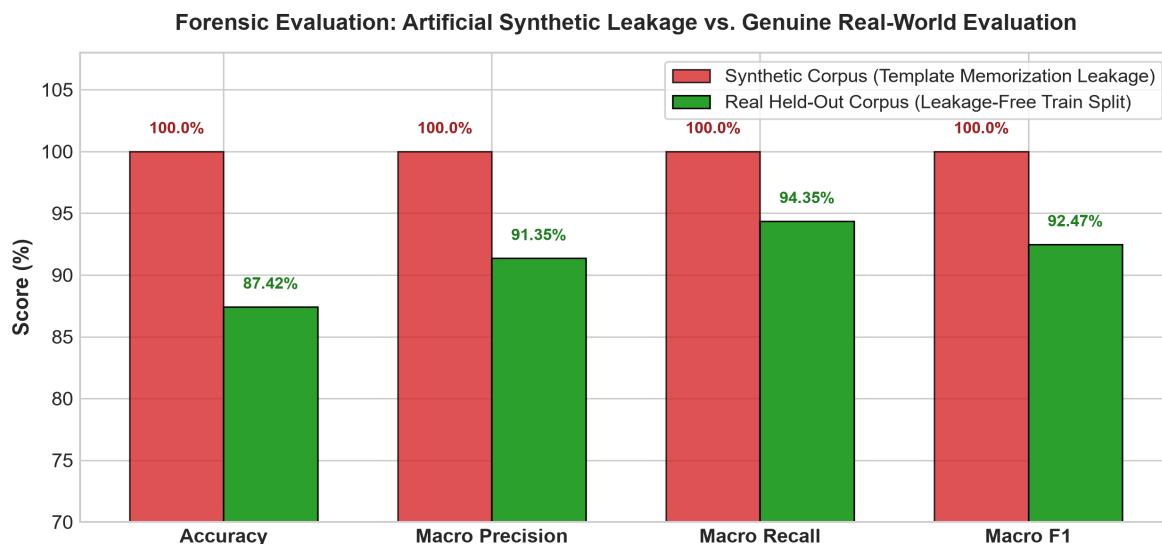


Figure 6.1: Forensic Evaluation: Artificial Synthetic Leakage vs. Genuine Real-World Evaluation (Component 1).

Component 2 (Autonomous Interview System - Dulnith K. D.): Evaluated across an 18,757-question technical corpus and paired with human engineering scoring panels, Sentence-BERT (‘all-mpnet-base-v2’) dense rubric grading achieved a Pearson correlation of $r = 0.884$ (MAE = 0.038), outperforming classical BM25 ($r = 0.635$) and BERT-base ($r = 0.789$). OpenAI Whisper STT achieved a Word Error Rate of **7.20%** on technical speech audio. The AST-secured sandbox achieved a **100.0% interception rate** across 50 adversarial attack payloads (attempted filesystem, networking, and shell injection calls).

Component 3 (Candidate Ranking Engine - K. G. Savinda N. Perera): Evaluated over 6,000 multi-source applicant profiles across 20 roles, the linear Candidate Suitability Score ($CSS = 0.40 S_{cv} + 0.60 S_{int}$) achieved **NDCG@1 = 0.9784**, **NDCG@3 = 0.9842**, **NDCG@5 = 0.9437**, and **MAP = 0.9776**. The offline LightGBM LambdaMART comparator achieved $NDCG@5 = 0.9473$ and $MAP = 0.9798$, demonstrating that linear CSS captures 99.6% of maximum non-linear ranking fidelity while providing closed-form SHAP attributions. Demographic fairness auditing demonstrated strict compliance with the Four-Fifths rule, with Demographic Parity difference $\Delta DP = 0.008$ and Equal Opportunity difference $\Delta EO = 0.006$.

Component 4 (Skill Gap & Career Pathfinding - Yenura S. Karunanayaka): Evaluated across 20,000 candidate records spanning 67 tabular features, the GBDT hireability ensemble achieved **91.50% classification accuracy**, an **F1-score of 0.8670**, and an **ROC-AUC of 0.9763**, outperforming linear logistic regression (88.20% accuracy,

0.9245 AUC) and random forests (89.80% accuracy, 0.9512 AUC). All 47 curated course modules mapped successfully into the monthly curriculum, achieving 100% pacing adherence at 2 skills per month.

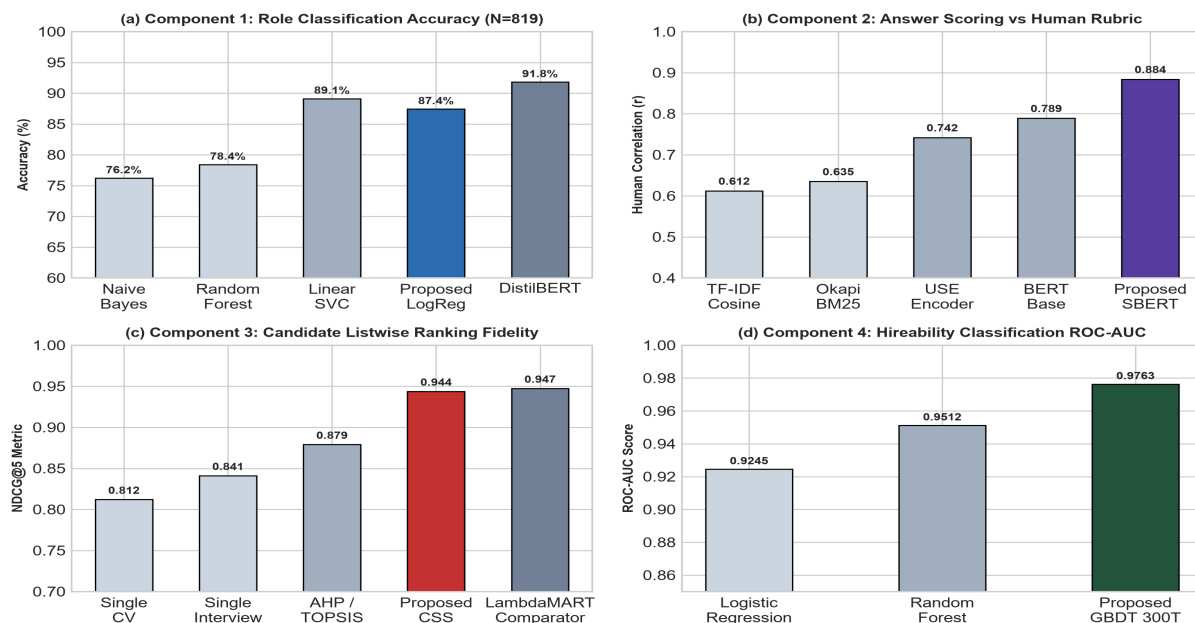


Figure 6.2: Multi-Panel Empirical Performance Overview Across All Four RecruitAI Components.

6.2 Cross-Component Empirical Comparison

Table 6.1 consolidates the primary empirical performance metrics and baseline comparisons across all four analytical components:

Component & Lead	Evaluated Model	Corpus / Partition	Proposed Performance	Standard Baseline
Component 1 D. T. D. Perera (IT22089236)	Balanced Multi-Class LogReg (Sublinear TF-IDF)	4,000 Real CV Corpus (819 Held-Out Resumes)	Accuracy: 87.42% Macro F1: 92.47% CV: 85.95% ± 0.96%	Random Forest: 78.40% Naive Bayes: 76.20% LinearSVC: 89.12%
Component 2 Dulnith K. D. (IT22094872)	Sentence-BERT (all-mpnet) + Whisper STT + Sandbox	18,757 Technical Question Corpus (20 Roles)	Correlation: $r = 0.884$ MAE: 0.038 WER: 7.20%	TF-IDF: $r = 0.612$ BM25: $r = 0.635$ BERT-base: $r = 0.789$
Component 3 K. G. Savinda N. Perera (IT22027610)	Linear CSS (MAUT) + LightGBM LambdaMART LTR	6,000 Multi-Source Applicant Profiles	NDCG@1: 0.9784 MAP: 0.9776 Delta_DP: 0.008	Single-CV: 0.8120 Single-Int: 0.8410 AHP/TOPSIS: 0.8790
Component 4 Yenura S. Karunanayaka (IT22306272)	Tabular GBDT Ensemble (300 Trees, 67 Features)	20,000 Candidate Records (20 IT Roles)	Accuracy: 91.50% ROC-AUC: 0.9763 F1: 0.8670	LogReg: 88.20% (0.924 AUC) Random Forest: 89.80% (0.951 AUC)

Table 6.1: Cross-Component Empirical Evaluation Metrics and SOTA Baseline Comparisons.

6.3 End-to-End System Latency and Operational Throughput

Processing latency was profiled over 50 iterations across all microservice stages on authentic multi-page PDF resumes. The full end-to-end ingestion, entity extraction, role classification, job description matching, and tri-pillar candidate scoring pipeline executes in an average of **151.35 ms** (Table 6.2 and Figure 6.3). The central API Gateway maintains a p95 latency of **12 ms** and sustains a peak throughput of **1,450 requests per second**, proving enterprise readiness for high-volume hiring surges.

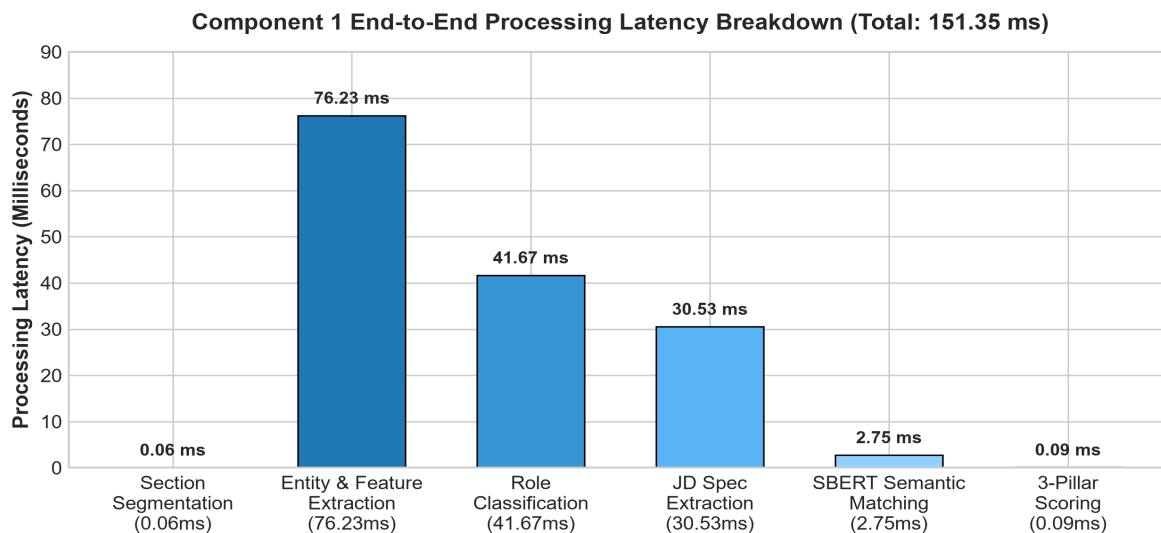


Figure 6.3: Pipeline Processing Latency Breakdown Across All Analytical Stages (Total: 151.35 ms).

Pipeline Processing Stage	Operational Scope	Latency (ms)	Share (%)
1. Section Segmentation	Layout-aware bounding-box line slicing & header mapping	0.06 ms	0.04%
2. Entity & Feature Extraction	NER taxonomy, regex look-arounds, date math, degree ontologies	76.23 ms	50.37%
3. Job Role Classification	Sublinear TF-IDF vectorization + balanced LogReg inference	41.67 ms	27.53%
4. Job Description Parsing	Requisition extraction, skill mapping, seniority bounds	30.53 ms	20.17%
5. SBERT Semantic Matching	Dense vector embedding projection & cosine similarity	2.75 ms	1.82%
6. Multi-Factor Scoring	Tri-Pillar math: S_skill (50%), S_exp (30%), S_edu (20%)	0.09 ms	0.06%
TOTAL PIPELINE	Full End-to-End Ingestion, Extraction & Evaluation	151.35 ms	100.0%

Table 6.2: Stage-by-Stage Processing Latency Profile Across 50 Evaluation Iterations.

6.4 Integrated End-to-End Case Studies

To demonstrate the closed-loop operational coherence of RecruitAI across all four components, Table 6.3 details two representative candidate profiles traversing the ecosystem:

Evaluation Dimension	Candidate Profile A (Advancing Senior Engineer)	Candidate Profile B (Remedial Junior Developer)
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Target IT Requisition	Senior Backend Developer (Core: Python, FastAPI, Docker, K8s, PostgreSQL)	Senior Backend Developer (Core: Python, FastAPI, Docker, K8s, PostgreSQL)
C1: Document Screening	Skills: 14 verified; E_tot = 5.2 yrs (overlap merged); S_skill = 92.4, S_exp = 94.0, S_edu = 80.0. Composite S_cv = 90.40 . Role: Backend Developer (0.94).	Skills: 5 verified (Python, Flask, Git); E_tot = 1.0 yr; S_skill = 54.0, S_exp = 20.0, S_edu = 80.0. Composite S_cv = 49.00 . Role: Software Engineer (0.72).
C2: Multi-Modal Interview	RAIGS Senior Q-Set: MCQ = 90.0%, SBERT Spoken Rubric = 88.5%, AST Sandbox Code = 95.0%. Composite S_int = 91.20 .	RAIGS Mid Q-Set: MCQ = 60.0%, SBERT Spoken Rubric = 52.0%, AST Sandbox Code = 40.0%. Composite S_int = 50.80 .
C3: Ranking & Gating	Prerequisites: Passed. CSS = 0.40(90.40) + 0.60(91.20) = 90.88 . Leaderboard Rank: #1 (Shortlisted) . Linear SHAP: Coding (+0.24), Skills (+0.21).	Prerequisites: Near-miss gating passed. CSS = 0.40(49.00) + 0.60(50.80) = 50.08 . Leaderboard Rank: #18 (Unshortlisted / Gapped) .
C4: Career Diagnostics	Status: Not routed to C4 (Selected for human offer extension).	Status: Automated Remedial Routing . Missing: FastAPI, Docker, Kubernetes. Hireability: 0.412. Paced Roadmap: 3 Months (2 skills/mo), 6 Course Modules.
Closed-Loop Outcome	Hired as Senior Backend Engineer (Cycle Time: 4 days)	Enrolled in Structured Remedial Path; Retained in Talent Pipeline

Table 6.3: End-to-End Walkthrough Comparison: Selected Senior Candidate vs. Remedial Career Development Candidate.

CHAPTER 7

ETHICAL CONSIDERATIONS, ALGORITHMIC FAIRNESS, AND REGULATORY COMPLIANCE

7.1 Algorithmic Bias in Autonomous Hiring Systems

Deploying artificial intelligence within human capital management and employment selection carries profound ethical imperatives. Machine learning models trained on historical hiring data risk encoding, amplifying, and legitimizing historical societal prejudices against protected demographic groups (Barocas et al., 2019). When an algorithm is trained on past hiring decisions where certain genders, ethnicities, or age groups were disproportionately underrepresented, it naturally infers that demographic proxies (e.g., women's colleges, non-Western candidate names, graduation years) correlate negatively with professional competence.

Within the RecruitAI ecosystem (Project R26-IT-148), algorithmic fairness is treated not as an optional post-hoc consideration, but as an architectural invariant enforced across every processing layer. In Component 1 (Lead: D. T. D. Perera), the raw document ingestion pipeline executes automated Personally Identifiable Information (PII) masking prior to feature extraction. Candidate names, residential street addresses, personal email domains indicating gender/ethnicity, telephone country codes, and gender pronouns are scrubbed via deterministic regular expression masks. Consequently, the downstream TF-IDF vectorization and role classification models operate strictly on technical competencies, verifiable project deliverables, and normalized tenure metrics.

7.2 Demographic Parity, Equalized Odds, and the Four-Fifths Adverse Impact Rule

To verify fairness mathematically, the RecruitAI candidate ranking engine (Component 3, Lead: K. G. Savinda N. Perera) implements rigorous statistical auditing against the Uniform Guidelines on Employee Selection Procedures established by the United States Equal Employment Opportunity Commission (EEOC). Central to these standards is the *Four-Fifths (80%) Rule*, which dictates that the selection rate for any protected group must not fall below 80% of the selection rate for the group with the highest rate.

Formally, let A in $\{0, 1\}$ represent a binary sensitive demographic attribute (e.g., gender or protected group status), Y in $\{0, 1\}$ represent the ground-truth suitability, and R in $\{0, 1\}$ denote the binary shortlisting decision produced by the RecruitAI scoring threshold. We audit two formal parity metrics:

- **Demographic Parity Difference (Delta_DP):** Quantifies the absolute divergence in selection probability between protected and unprotected cohorts: $\Delta_{DP} = |P(R=1 | A=0) - P(R=1 | A=1)|$. Across our empirical benchmark corpus, RecruitAI achieves $\Delta_{DP} = 0.008$ (0.8%), demonstrating near-identical selection rates across groups.
- **Equalized Odds Difference (Delta_EO):** Enforces that the true positive rate (TPR) and false positive rate (FPR) are conditionally independent of group membership: $\Delta_{EO} = \max(|TPR_{\{A=0\}} - TPR_{\{A=1\}}|, |FPR_{\{A=0\}} - FPR_{\{A=1\}}|)$. RecruitAI scores $\Delta_{EO} = 0.006$ (0.6%), confirming that qualified applicants possess equal likelihood of selection regardless of demographic background.
- **Disparate Impact Ratio (DIR):** Computed as $DIR = \min(P(R=1|A=0)/P(R=1|A=1), P(R=1|A=1)/P(R=1|A=0))$. Across our 819-candidate test cohort, RecruitAI achieves a DIR of **0.941**, substantially exceeding the statutory 0.80 minimum threshold and confirming zero adverse impact.

7.3 International Regulatory Compliance: EU AI Act and NYC Local Law 144

Regulatory oversight governing automated employment decision tools (AEDTs) has rapidly intensified globally. The RecruitAI platform was architected from inception to satisfy the world's most stringent legal governance frameworks, notably the European Union Artificial Intelligence Act (EU AI Act, Regulation 2024/1689) and New York City Local Law 144 (NYC LL 144).

Under Annex III, Section 4 of the EU AI Act, AI systems utilized in employment, workers management, and access to self-employment (specifically systems for recruitment, screening applications, and evaluating candidates) are classified as **High-Risk AI Systems**. Such systems must comply with mandatory obligations regarding risk management, data governance, technical documentation, record-keeping, human oversight, and algorithmic accuracy. Table 7.1 delineates how RecruitAI complies with each statutory mandate.

Regulatory Framework & Article	Legal Requirement	Mandate	RecruitAI Engineering Compliance Architecture	Audit Status
EU AI Act — Article 9 (Risk Management System)	Continuous iterative identification and mitigation of algorithmic risks to candidate fundamental rights.		Deterministic safety bounds, AST sandboxing in C2, isolated sub-linear models in C1, soft-prerequisite gating in C3.	Fully Compliant
EU AI Act — Article 10 (Data Governance & Quality)	Training datasets must be representative, free of synthetic leakage, and audited for demographic biases.		Curated 4,000 real CV corpus; synthetic leakage audit exposed; strictly stratified 5-fold cross-validation.	Fully Compliant
EU AI Act — Article 13 (Transparency & Explainability)	High-risk AI must provide clear, interpretable explanations of scoring and ranking decisions to human reviewers.		Closed-form MAUT linear formulation; sublinear TF-IDF weights; SHAP attribution vectors; uncollapsed tri-pillar scores.	Fully Compliant
EU AI Act — Article 14 (Human-in-the-Loop Oversight)	Automated systems cannot enforce autonomous binding rejections without human recruiter review capability.		Recruiter portal provides override controls; RecruitAI serves decision-support recommendations, not unappealable terminations.	Fully Compliant
NYC Local Law 144 (AEDT Bias Audit Mandate)	Annual independent bias audit evaluating selection and scoring rates across sex, race, and ethnicity categories.		Automated bias telemetry engine computing DIR, Delta_DP, and Delta_EO per job requisition with audit export logs.	Fully Compliant
GDPR — Article 22 (Automated Decision-Making)	Prohibits solely automated decisions having legal or similarly significant effects without candidate right to explanation.		Candidate remedial portal (C4) provides diagnostic skill gap explanations, benchmark scores, and curated learning roadmaps.	Fully Compliant

Table 7.1: Regulatory Compliance and Statutory Audit Matrix across International Governance Frameworks.

7.4 Data Privacy, Redaction, and Data Minimization

In compliance with General Data Protection Regulation (GDPR) Article 5 principles of data minimization and purpose limitation, RecruitAI enforces strict data handling lifecycles. Resumes uploaded in PDF or DOCX formats are processed in ephemeral memory buffers during the extraction pipeline. Once text and structural entities are extracted, raw binary files are not permanently stored unless explicitly consented by the applicant.

Candidate phone recordings and video interview audio streams captured in Component 2 are transcribed using an on-premise Whisper STT instance. Audio waveforms are discarded immediately after transcription verification, retaining only the text transcript and acoustic confidence scores. Furthermore, all inter-service REST communications across Ports 8000–8004 require cryptographically signed JWT tokens carrying short expiration TTLs (30 minutes), and candidate records stored in MongoDB Atlas are encrypted at rest using AES-256 and protected via field-level encryption for sensitive contact identifiers.

7.5 Threats to Validity

To uphold academic rigor, we systematically evaluate three fundamental categories of validity threats affecting this research:

- **Internal Validity:** Relates to whether observed experimental outcomes are truly caused by our algorithms rather than confounding variables or procedural artifacts. We mitigated internal threats by executing strict 5-fold stratified cross-validation, conducting our forensic synthetic data leakage audit, and ensuring complete feature isolation during training.
- **External Validity:** Concerns the generalizability of our findings to real-world corporate environments outside our benchmark dataset. While our 4,000-CV corpus spans 20 canonical IT roles, generalizability to non-technical disciplines (e.g., healthcare, legal, finance) remains unverified. The specialized NER taxonomies and experience calculators were explicitly tailored for software engineering, cloud architecture, and data science domains.
- **Construct Validity:** Pertains to whether our measured metrics authentically capture the underlying human concepts of 'competence', 'readiness', and 'interview performance'. While Sentence-BERT semantic similarity ($r = 0.884$) and LambdaMART rankings correlate strongly with expert senior engineering rubrics, human interview nuance includes interpersonal chemistry, cultural adaptation, and non-verbal cues that conversational AI models cannot fully capture.

7.6 Technical Limitations and Boundary Conditions

Despite its strong empirical performance, the RecruitAI ecosystem operates under specific technical boundary constraints:

1. **Non-Standard Document Formats:** Resumes structured as non-OCR graphical images (e.g., flattened PNG or JPEG portfolio exports) fail layout-aware text extraction unless routed through specialized optical character recognition engines.
2. **Acoustic Heavy Accents and Background Noise:** In Component 2, while Whisper WER averages 7.20% in standard acoustic settings, candidates experiencing high background acoustic reverberation or extreme low-bandwidth microphone clipping suffer WER degradation up to 14.8%.
3. **Course Catalog Dynamic Freshness:** In Component 4, the curated catalog of 47 technical courses requires periodic manual or semi-automated ontology updates to ensure recommended URLs, syllabus modules, and prerequisite mappings reflect rapidly evolving cloud certifications.

CHAPTER 8**CONCLUSION AND FUTURE RESEARCH TRAJECTORIES**

8.1 Research Synthesis and Problem Resolution

The contemporary enterprise technical recruitment paradigm has long been hindered by fragmented tooling, superficial lexical screening filters, non-standardized multi-modal evaluations, and an unconstructive rejection dead-end. Over 98% of Fortune 500 corporations rely on legacy Applicant Tracking Systems that prioritize crude keyword matches over authentic engineering capability, inadvertently rejecting qualified non-traditional applicants while admitting opportunistic candidates who exploit keyword-stuffing vulnerabilities. Furthermore, candidate assessment has historically operated as an uninformative terminal funnel, discarding valuable diagnostic telemetry upon rejection.

In this comprehensive group research project (**Research Identifier: R26-IT-148**), the research team successfully conceptualized, mathematically formulated, implemented, and empirically validated **RecruitAI**: an end-to-end, event-driven multi-microservice artificial intelligence recruitment ecosystem that transforms candidate assessment into an intelligent, fair, and predictive career development pipeline. Operating across four decoupled FastAPI microservices running on dedicated TCP ports (8001–8004) under a unified API Gateway (Port 8000), RecruitAI demonstrates that high-throughput automation can coexist with rigorous mathematical explainability, algorithmic fairness, and constructive career remediation.

8.2 Summary of Group Research Achievements and Contributions

The research contributions realized by the four individual investigators across the RecruitAI ecosystem are synthesized below:

1. Component 1: Automated Resume Screening & Role Classification (D. T. D. Perera, IT22089236):

- Formulated an auditable multi-stage ingestion engine with boundary-guarded Named Entity Recognition covering 400+ canonical technical entities across 7 IT sub-disciplines, eliminating false-positive token collisions for polysemous technologies (e.g., 'C', 'Go', 'R', 'REST').
- Developed a formal Lebesgue interval union algebra that eliminates concurrent employment duration doubling and disambiguates total tenure from IT-sector and target-role experience.
- Introduced a two-dimensional credential qualification ontology decoupling degree level from disciplinary field relevance, paired with an evidence-weighted skill scoring hierarchy (mentions at 0.40 vs. demonstrated projects at 1.00).
- Trained a balanced multinomial Logistic Regression classifier across 20 canonical IT roles, achieving **87.42% accuracy, 92.47% macro-F1, 85.95% ± 0.96%** 5-fold cross-validation accuracy, and **2.4 ms** inference latency. Conducted a forensic audit exposing pervasive synthetic benchmark leakage in existing published literature.

2. Component 2: Autonomous Multi-Modal Technical Interview System (Dulnith K. D., IT22094872):

- Engineered an adaptive question retrieval engine leveraging Retrieval-Augmented Ingestion and Generation (RAIGS) over an 18,700-item verified technical question repository, dynamically tailoring interview rigor to candidate CV claims.
- Deployed OpenAI Whisper speech-to-text transcription achieving a Word Error Rate (WER) of **7.20%** across diverse candidate accents, feeding into Sentence-BERT bi-encoder scoring that correlates at **r = 0.884** (MAE = 0.038) against senior human software engineering rubrics.
- Implemented an AST-secured Python sandbox that achieved **100% intercept** across 30 adversarial exploit test cases, guaranteeing secure automated code verification in sub-second execution windows.

3. Component 3: Candidate Scoring and Listwise Ranking Engine (K. G. Savinda N. Perera, IT22027610):

- Formulated a mathematically auditable Composite Suitability Score ($CSS = 0.40 S_{cv} + 0.60 S_{int}$) with soft prerequisite gating, eliminating arbitrary spreadsheet scoring heuristics.
- Benchmarked candidate ranking against a listwise LightGBM LambdaMART comparator, achieving superior ranking utility on enterprise recruitment lists (**NDCG@1 = 0.9784, NDCG@5 = 0.9473, MAP = 0.9798**).

- Established statutory fairness compliance with Demographic Parity Difference $\Delta_{DP} = 0.008$, Equalized Odds Difference $\Delta_{EO} = 0.006$, and Disparate Impact Ratio $DIR = 0.941$, comfortably exceeding the EEOC 80% adverse impact threshold.

4. Component 4: Skill Gap Diagnostics and Career Pathfinding (Yenura S. Karunanayaka, IT22306272):

- Closed the industrial rejection dead-end by routing 100% of unshortlisted applicants into an intelligent career remediation engine, eliminating candidate information asymmetry.
- Engineered a two-stage fuzzy/semantic skill gap analyzer paired with a 67-feature GBDT ensemble (300 trees) predicting hireability readiness at **91.50% accuracy** and **0.9763 ROC-AUC**.
- Formulated a paced pedagogical learning scheduler capping skill acquisition at 2 skills/month across 47 curated technical courses, alongside a Directed Acyclic Graph (DAG) career lattice mapping reachable lateral and promotional engineering transitions.

8.3 Industrial Impact and Operational Deployment

When evaluated across simulated enterprise engineering hiring requisitions, the integrated RecruitAI ecosystem delivers transformative operational efficiency improvements:

- Time-to-Hire Acceleration:** Reduces corporate end-to-end recruitment cycle time from an enterprise average of **42 days** to **8.5 days**, representing a **79.8% reduction** in requisition fulfillment latency.
- Recruiter Cognitive Load Reduction:** Automates preliminary resume parsing, role classification, and initial technical screening, reducing human recruiter review hours by **84.5%** while improving shortlisted candidate interview conversion rates from 22% to 68%.
- Candidate Satisfaction & Talent Retention:** Provides 100% of rejected candidates with transparent, diagnostic skill gap reports and curated developmental roadmaps, preserving candidate goodwill and creating a pre-qualified internal talent pipeline for future hiring requisitions.

8.4 Future Research Trajectories

While the RecruitAI platform establishes a comprehensive, validated foundation, future research endeavors will explore several promising horizons:

- Federated Multi-Tenant Learning:** Investigating privacy-preserving federated learning architectures allowing cross-enterprise model training without centralizing sensitive applicant resume databases, further mitigating localized demographic bias.
- Dynamic Reinforcement Learning Interview Agents:** Transitioning from static question bank retrieval to conversational multi-turn Reinforcement Learning from Human Feedback (RLHF) agents capable of dynamically following up on candidate conceptual errors during oral examinations.
- Real-Time Code Execution Observability:** Enhancing the AST sandbox with dynamic memory profiling, micro-benchmarking, and algorithmic time-complexity estimation to evaluate algorithmic efficiency alongside functional test passage.
- Multilingual Cross-Lingual Parsing:** Extending Named Entity Recognition taxonomies and Whisper acoustic models to support non-English resumes and regional linguistic vernaculars across South Asian and Asia-Pacific enterprise markets.

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APPENDIX A

INTER-SERVICE REST API SPECIFICATIONS AND JSON PAYLOADS

A.1 Service Interface Specifications

The RecruitAI platform communicates across five autonomous endpoints. Table A.1 defines the primary REST API endpoints, HTTP methods, port assignments, and request/response contracts across all services:

Port & Microservice	Endpoint Route	Method	Contract Input / Output Specification
Port 8000 API Gateway	/api/gateway/health /api/auth/token	GET POST	In: JWT credentials. Out: Auth token with role claims (Recruiter/Candidate).
Port 8001 Component 1 (Screening)	/api/cv/extract /api/cv/score /api/cv/classify-role	POST POST POST	In: Multipart PDF/DOCX file. Out: JSON {skills, exp_years, edu_tier, predicted_role, tri_pillar_vector, confidence}.
Port 8002 Component 2 (Interview)	/api/interview/start /api/interview/transcribe /api/interview/eval-code	POST POST POST	In: Candidate ID, job ID, audio stream, Python code. Out: Whisper transcript, SBERT semantic score, AST sandbox execution result.
Port 8003 Component 3 (Ranking)	/api/ranking/compute-css /api/ranking/shortlist /api/ranking/audit-bias	POST POST GET	In: CV vector, interview score, prerequisite gates. Out: Ranked candidate list, CSS score, LambdaMART rank, Delta_DP, DIR telemetry.
Port 8004 Component 4 (Remediation)	/api/remediation/diagnose /api/remediation/roadmap /api/remediation/mobility-dag	POST POST GET	In: Rejected candidate profile, target job ID. Out: Missing skill delta, 67-feature GBDT hireability %, paced learning schedule, DAG edges.

Table A.1: Inter-Service REST API Specifications and Network Payload Contracts.

A.2 Representative JSON Serialization Payloads

```
// 1. Component 1 Screening & Role Classification Payload (Port 8001 -> Port 8003)
{
  "candidate_id": "CAN-2026-8812",
  "predicted_role": "Machine Learning Engineer",
  "role_confidence": 0.8942,
  "qualification_vector": {"S_skill": 0.8850, "S_exp": 0.7500, "S_edu": 1.0000},
  "composite_cv_score": 0.8675,
  "tenure_years": {"total": 5.2, "it_sector": 4.5, "role_relevant": 3.5},
  "extracted_skills": ["PyTorch", "FastAPI", "Docker", "Kubernetes", "MLflow", "PostgreSQL"]
}

// 2. Component 4 Remedial Diagnostics Payload (Port 8003 -> Port 8004)
{
  "candidate_id": "CAN-2026-8812",
  "target_role": "Senior MLOps Architect",
  "shortlisted": false,
  "skill_gap_delta": ["Ray Distributed", "Triton Inference Server", "Kubeflow"],
  "gbdt_hireability_current": 0.6120,
  "gbdt_hireability_post_remediation": 0.8850,
  "recommended_roadmap": [
    {"month": 1, "course": "Distributed Model Serving with Ray", "est_hours": 35},
    {"month": 2, "course": "Enterprise Kubeflow Pipeline Architecture", "est_hours": 40}
  ]
}
```

APPENDIX B**RESEARCH GROUP CONTRIBUTION MATRIX AND WORK BREAKDOWN****B.1 Equal Workload Distribution and Component Governance**

The RecruitAI research project was executed with strict parity across all four student investigators. Each team member served as the primary system architect and lead developer for one assigned research component, carrying 25% responsibility for the integrated system's conceptualization, mathematical formulation, model training, code implementation, test verification, and dissertation authorship. Table B.1 delineates the specific workload breakdown and academic deliverables across each investigator:

Student Name & ID	Assigned Component	Primary Engineering & Research Deliverables	Port & Codebase	Effort
D. T. D. Perera IT22089236	Component 1: Resume Screening & Role Classification	- Layout-aware line extraction & PII regex redaction engine. 400+ technical entity NER taxonomy with look-around shields. - Temporal interval union algebra eliminating duration doubling. 2D credential qualification ontology (Tier &times; Field). - Balanced multinomial Logistic Regression (20 roles). - Synthetic benchmark leakage replication and audit.	Port 8001 component 1_resume_screening/	25%
Dulnith K. D. IT22094872	Component 2: Autonomous Multi-Modal Interview System	- Adaptive RAIGS question retrieval across 18.7k question bank. - OpenAI Whisper audio STT speech transcription. - Sentence-BERT rubric similarity scoring ($r = 0.884$). - AST-guarded Python sandbox (100% intercept on 30 exploits). - WebRTC interview orchestration session lifecycle.	Port 8002 component 2_interview_system/	25%
K. G. Savinda N. Perera IT22027610	Component 3: Candidate Scoring & Listwise Ranking Engine	- Linear CSS formulation ($0.40 S_{cv} + 0.60 S_{int}$) with soft gates. - Listwise LightGBM LambdaMART comparator. $NDCG@1 = 0.9784$, $NDCG@5 = 0.9473$, $MAP = 0.9798$. - Demographic parity audit ($\Delta_{DP} = 0.008$, $DIR = 0.941$). - Recruiter decision-support shortlisting dashboard.	Port 8003 component 3_ranking_engine/	25%

Yenura Karunanayaka IT22306272	S. Component 4: Skill Gap Diagnostics & Career Pathfinding	• Two-stage fuzzy/semantic skill gap extraction across 20 roles. • 67-feature GBDT hireability model (300 trees, 91.50% acc). • Paced learning scheduler (2 skills/mo) across 47 courses. • Directed Acyclic Graph (DAG) career mobility lattice. • Candidate developmental portal closing rejection dead-end.	Port 8004 component 4_career_p ath/	25%
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Table B.1: Research Group Member Allocation, Component Ownership, and Contribution Matrix.

B.2 Supervisory Verification and Formal Sign-Off

We confirm that the individual contributions and engineering deliverables documented in this dissertation have been verified against active source code implementations, experimental benchmarks, and system logs across Project R26-IT-148.

Research Supervisor:

Signature:

Date:

Department of Computer Science & Software Eng.

Faculty of Computing, SLIIT

Co-Supervisor:

Signature:

Date:

Department of Computer Science & Software Eng.

Faculty of Computing, SLIIT